

Legislator ex Machina

Simulating Parliamentary Discourse using LLMs

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Git-Repository <https://github.com/jelenameyer/dlss-project-lem>
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1 Introduction

This report investigates the extent to which large language models (LLMs) can authentically replicate parliamentary discourse patterns of German political parties through fine-tuned speech generation. We develop two complementary models: a generative LLM trained to produce party-specific political speeches, and a classification LLM that discriminates between speeches from six major German parties. Model validation employs three methodologies: automated evaluation via the classifier (constituting a political Turing test), and systematic comparison of generated responses to survey questions against empirical party positions from Wahl-O-Mat data, and a systematic analysis of the linguistic and rhetorical features of generated texts relative to authentic speeches. Using speeches from the 19th and 20th German Bundestag, we assess whether contemporary LLMs can capture the nuanced rhetorical signatures that distinguish party discourse.

2 Related Work and Theoretical Context

Political speeches have been the subject of various LLM analyses. Doan (2024) showed, how political views could be extracted from political speeches in Norway, while Le Mens and Gallego (2025) performed similar analyses for the US, UK and the EU. Chalkidis and Brandl (2024) used Llama Chat to imitate political decisions of national parties in the European parliament. Heyde, Haensch, and Wenz (2025) predicted survey behavior with data from US-American and German voters. Beyond this, the work by Kato et al. (2024) showed, how political views in speech data are a potent data base for LLMs to generate realistic speeches. This was also done for speeches in US-presidential debates by Z. Liu et al. (2024).

We aim to continue along this line of research by combining existing approaches. Our goal is to generate political speeches based on knowledge derived from past speeches (see Z. Liu et al. (2024)), but using data from Germany (see Heyde, Haensch, and Wenz (2025)). Z. Liu et al. (2024) employed GPT-4 and Claude as their models. Our work aligns with previous studies, such as Savoy (2024), who generated political speeches for U.S. presidents using GPT-4, and D. Labbé, C. Labbé, and Savoy (2025), who applied the same LLM to French presidents. After contacting Savoy about his methodology, we learned that he only prompted GPT to write speeches, but did not train the model on speech data. Since we lack financial funding, we will rely on public accessible Llama models for our analysis. Specifically, we compare the performance of two Llama generations: a Llama-2 model with 13 billion parameters¹ and a Llama-3 model with 8 billion parameters². Both models are instruction-based language models, optimized for the German language, have a manageable size and can hence be fine-tuned for generating political speeches as demonstrated by Chalkidis and Brandl (2024). In past studies, validation checks were done by either employing another LLM to check for congruency between generation and ground truth (see Z. Liu et al. (2024)) or by conducting a validation survey like Herbold et al. (2024) did. The authors asked respondents to rate real and artificial statements concerning authenticity, coherence and relevance. Over all three metrics, the respondents preferred the LLM statement over the true statement.

We will combine those approaches and apply a three-fold validation. First, we build a BERT-based validation model, which will classify the generated speeches, and reveal if they pass as real party speeches. Afterwards, we are going to use existing survey questions (from the Wahl-O-Mat) to measure congruence between the party representation our models learned and true survey answers. Finally, we will conduct the same type of semantic analysis as Savoy (2024), by applying the LIWC dictionary to evaluate the extent to which our generated speeches semantically and rhetorically align with real political speeches.

¹Llama-3-13b-chat-german

²Llama3-DiscoLeo-Instruct-8B-v0.1

Despite their strengths in generating and classifying political speech, LLMs show notable biases. Heyde, Haensch, and Wenz (2025) find that models often fail to represent the full political spectrum, showing a skew toward green and left-leaning positions. Similar ideological biases are reported by Z. Liu et al. (2024) and Batzner et al. (2024). On the other hand, a study by Agiza, Mostagir, and Reda (2024) demonstrates, that fine-tuning can shift a model’s political stance. This suggests that political neutrality is not inherent but can be adjusted through targeted training. These potential sources of error will be monitored and taken into account in the methods and results.

Overall, we contribute to the described field of research by applying a two-stage LLM approach where we combine a generation and classification model. Furthermore, we expand the field of studied languages by working with German parliamentary speeches.

3 Data Analysis

3.1 Data Collection

Our dataset comprises speeches from the 19th Bundestag (October 2017 - October 2021) and 20th Bundestag (October 2021 - February 2025), retrieved via Bundestag-Mine³. We decided to use the Bundestags-Mine as our data source, because this platform provides the Bundestag session protocols as pre-processed JSON files, which contain both the speeches as continuously formatted strings and metadata such as legislature, session, protocol number, agenda item title and date of the session. Speaker metadata including party affiliations were obtained through the Bundestag-Mine API endpoint⁴ and merged with speeches using their unique speaker identifiers.

We supplemented this with party manifestos from the Manifesto Data Project⁵ and official survey data from the 2025 Bundestagswahl Wahl-O-Mat dataset⁶. All datasets were standardized to CSV format, with speeches linked to speakers and metadata through respective ID mappings.

3.2 Preprocessing

The initial dataset contained 44,716 speeches. The following preprocessing steps were applied:

First, we cleaned the data. Duplicate detection revealed no duplicate entries. Missing value analysis revealed that 60 speeches were lacking a speaker party. Since the number was low compared to the overall number of speeches in our data, we dropped those speeches.

The linked meta-information on speeches helped to filter for meaningful political speeches. A manual look through the speech titles identified a set of keywords as indicators for a procedural speech such as “election” and “rules of procedure”⁷. We excluded those speeches using keyword filters on the speech titles, removing 6,352 entries.

Length-based filtering eliminated speeches below 200 words (roughly the 5th percentile) and above 1,500 words (approx. 99th percentile), targeting speeches that were outliers in length and therefore probably unhelpful as training data while only excluding 1,771 speeches from the analysis. Short speeches were mainly answering specific questions while long speeches were mainly from the chancellors and therefore unrepresentative for a specific party. We also excluded minor parties (PLoS, LKR, Die PARTEI, SSW) with insufficient representation (416 speeches), retaining the six major parties. CDU and CSU were combined as a single entity reflecting their parliamentary alliance. Several party names were shortened for easier processing. Finally, politician names were combined into one column for the full name, and auxiliary columns removed.

3.3 Exploratory Findings

The dataset includes the speech text and metadata such as speaker name, party affiliation, legislative period, date, and speech title.

Our final dataset contains 36,117 speeches with a vocabulary of 319,779 unique tokens. Speech length exhibits consistent left-skewed distributions across all parties (Figure 1), with mean length of 634 words (median 607), indicating a predominance of shorter contributions with a long tail of lengthier speeches.

The dataset exhibits substantial class imbalance reflecting parliamentary composition: CDU/CSU dominates with 26.8% of speeches, followed from the SPD with 22.2%, GRÜNE with 14% and FDP with 13%. THE LINKE accounts for the fewest speeches at only 10.0% (Figure 2, top).

³Bundestag-Mine

⁴API: <https://bundestag-mine.de/api/DashboardController/GetSpeakerById/{speakerId}>.

⁵Manifesto Data

⁶Wahl-O-Mat Data from Bundeszentrale für politische Bildung

⁷Wahl, Eröffnung, Eidesleistung, Bekanntgabe, Fragestunde, Befragung, Nationalhymne, Geschäftsordnung

This mirrors the parties' relative parliamentary strength but poses potential challenges for balanced model training. We retained the natural distribution to preserve maximum data, evaluating balancing strategies during fine-tuning.

Overall, 53% of speeches occurred in the 19th legislative period and 47% in the 20th. Temporal analysis reveals fluctuations in party speech shares corresponding to changing government majorities (Figure 2, bottom). CDU/CSU, the largest faction in the 19th Bundestag, held the highest speech share initially. From 2022 onward, SPD and CDU/CSU maintain the highest shares, reflecting the transition in governing coalitions. Other parties' speech proportions remained relatively stable.

Speech distribution varies substantially at the individual level, with Volker Ullrich (CSU) delivering the most speeches, followed by Helge Lindh (SPD). This concentration suggests certain members of parliament (MPs) serve as primary party spokespeople, potentially introducing speaker-specific stylistic patterns that may confound party-level classification. Given our large dataset, we did not correct for this phenomenon.

Text preprocessing for all lexical analyses involved lowercasing, punctuation removal, and tokenization. After removing German stopwords using a German stopwords list and hand-picked parliamentary procedure terms⁸, the most frequent terms -'euro', 'land', 'antrag'- reflect core policy domains.

Topic analysis of speech titles revealed concentrated debate around foreign affairs and economic policy (Table 1). Critically, comparing common topics and words across parties showed minimal differentiation - all six parties exhibit nearly identical frequent terms and topics with only minor ordering differences. This suggests LLMs cannot rely on simple topical or lexical features alone for party classification, necessitating more sophisticated pattern recognition.

Further exploratory details, including word clouds per party, are available in the accompanying Notebook⁹.

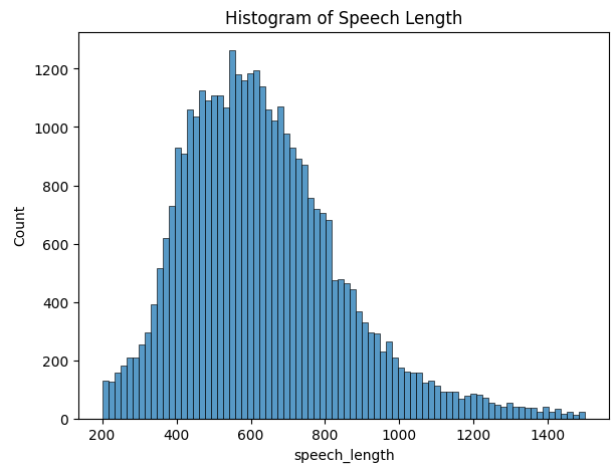


Figure 1: Speech length distribution

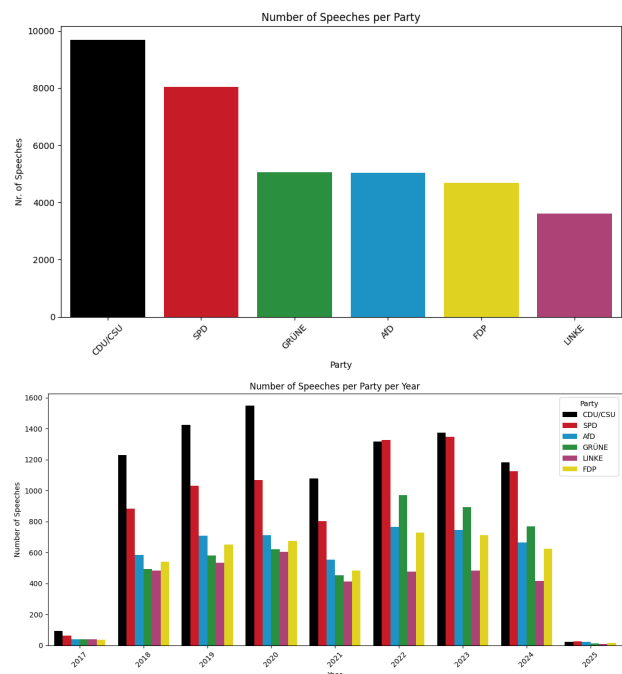


Figure 2: Number of speeches per party and speeches over time

⁸Parliamentary Procedure terms: "damen", "herren", "herr", "kollegen", "kolleginnen", "menschen", "deutschland", "liebe", "präsident", "präsidentin"

⁹Jupyter Notebook: 1.1.EDA.ipynb

#	Topic Title	Count
1	Auswärtiges Amt	275
2	Verteidigung	273
3	Wirtschaftliche Zusammenarbeit und Entwicklung	262
4	Abschließende Beratungen ohne Aussprache	262
5	Gesundheit	254
6	Ernährung und Landwirtschaft	228
7	Bildung und Forschung	214
8	Familie, Senioren, Frauen und Jugend	197
9	Überweisungen im vereinfachten Verfahren	179
10	Arbeit und Soziales	171
11	Digitales und Verkehr	139
12	Umwelt, Naturschutz und nukleare Sicherheit	119
13	Bundeskanzlerin und Bundeskanzleramt	115
14	Umwelt, Naturschutz, nukleare Sicherheit und Verbraucherschutz	113
15	Bundeswehreinsatz in Kosovo (KFOR)	110

Table 1: Top 15 full topic titles and their frequency

4 Methods

4.1 Political speech classification

The task of labelling political speeches with the correct party-authorship comes with several challenges:

Model choice: Since the task requires no text generation, an encoder model seemed appropriate and resource-efficient. We chose a base BERT model pretrained for the German language to save training costs and ensure adequate language comprehension. The model was adapted for classification using HuggingFace’s ‘AutoModelForSequenceClassification’, which is supported by the *bert-base-german-cased* model (bertGc). BERT’s deterministic, non-generative architecture produces stable, interpretable outputs without the variability seen in other LLMs, making it well-suited for classification tasks. We also experimented with *gbert-base* (gbert), for which the same arguments apply. Released in October 2020, Gbert is slightly larger than bertGc (111M vs. 110M parameters) and was trained on more recent German language data and corpora.

Dealing with lengthy input: BERT has a context window limit of 512 tokens, but most speeches exceed this length. While longformer models exist for longer texts, we found no adequately pretrained German variant. We instead used a sliding window approach during finetuning and inference: inputs were split into overlapping chunks, outputs were averaged to derive one label per speech.

Dealing with unbalanced data: As shown in Figure 2, the number of speeches per party varies greatly. Experiments with under-sampling revealed that the unbalanced dataset actually performed better. Additionally, we applied custom initial class weights for some models, as detailed in Chapter 4.1.3 Training.

Finetuning strategy: We chose to finetune the full BERT encoder and classifier head rather than using LoRa on attention layers only, as full finetuning showed significantly better results and was feasible given our resources.

Improving performance beyond finetuning: We combined two strategies: selected hyperparameter tuning and advanced data preprocessing. The latter involved “boiling down” speeches to their core content, described in detail in the following chapter.

4.1.1 Classification specific preprocessing

In preparation for the classification task, we converted the full dataset into the Hugging Face dataset format. We then split the data into training (70%), validation (15%), and test (15%) subsets.

To prevent data leakage, we removed all sentences from speeches that contained the name of the speaker’s own party. In proportion to the number of speeches per party, LINKE and AfD had the most self-mentions. It has to be noted that “linke” is a regular german word and there might be an “over-match” of results in that the word in its party-un-affiliated sense got detected and removed as well. The least party self mentions were found for the CDU/CSU and FDP.

In addition, we created an alternative data set (data version B) in which we removed German stop words and common parliamentary vocabulary. This gives us a string for each speech that is less coherent in terms of content but more focused in terms of political stance.

4.1.2 Models

We started with a simple setup and iteratively adjusted our approach in search of better performance. Our first model was *google-bert/bert-base-german-cased*¹⁰ (bertGc), trained on German Wikipedia, OpenLegalData, and news articles, last updated in April 2020. In our initial run, we finetuned all parameters using the standard 512-token context window. In the second attempt, we applied the sliding-window approach to include the full texts during finetuning and evaluation. We then performed hyperparameter tuning, though the original sliding-window configuration yielded the best results. The same model was also trained on the distilled dataset B, again with hyperparameter tuning. Next, we added initial class weights to mitigate class imbalance - returning to dataset A. Finally, we used *deepset/gbert-base*¹¹ (gbert) as a second BERT-based model for comparison. Incorporating learnings from earlier runs, we trained it on full texts, ran a version that used balancing class weights and went on to finetune hyperparameters for the weighted version - again on both datasets - as it performed better than the unbalanced one.

In total, we trained 7 model variants. Their configurations are listed in Table 2.

4.1.3 Training

Except for the vanilla model (2.1), we used a sliding window approach for all the models, having a window length of 512 and a stride of 256. Fine-tuning (models 2.2a/b and 2.5a/b) included learning rates (1e-5, 2e-5, 3e-5)¹², train batch sizes (8, 16, 32) training epochs (3 or 4) and weight decay (0, 0.01, 0.05). For BertGc we tested 12 combinations, for Gbert resources allowed to test 16 different configurations - the best can be seen in Table 2 for the respective models. For balancing, we overrode the default loss computation in HuggingFace's Trainer by implementing a custom WeightedTrainer class. This allowed us to apply predefined class weights directly in the cross-entropy loss. "Bal." for balanced in Table 2 indicates whether balancing has been performed or not. Evaluation of training occurred after every epoch on a held-out validation set, as loss function we applied cross entropy loss.

Name	Model	lr	Batch	Epochs	WD	Bal.	Dataset	Note
2.1 BERT-Vanilla	bertGc	2e-5	8 / 8	3	0.01	no	A	baseline, 512 token window
2.2 BERT-Fulltext	bertGc	2e-5	8 / 16	3	0.01	no	A	
2.2a BERT-TuneA	bertGc	2e-5	16 / 16	4	0.01	no	A	ntrials = 12
2.2b BERT-TuneB	bertGc	2e-5	8 / 16	3	0.05	no	B	ntrials = 12
2.3 BERT-Weighed	bertGc	2e-5	16 / 16	3	0.01	yes	A	
2.4 GBERT-Fulltext	gbert	2e-5	16 / 16	3	0.01	no	A	baseline GBERT, full text
2.5 GBERT-Weighed	gbert	2e-5	16 / 16	3	0.01	yes	A	
2.5a GBERT-TuneA	gbert	2e-5	16 / 16	4	0.01	yes	A	ntrials = 16
2.5b GBERT-TuneB	gbert	2e-5	8 / 16	3	0.05	yes	B	ntrials = 16

Table 2: Details on BERT-based trained models

4.1.4 Evaluation

We compared all fine-tuned models using accuracy, balanced accuracy (to address class imbalance), and macro-averaged F1-scores ("macro", to account for class imbalance). All fine-tuned model variants were evaluated on the validation set. Their validation performance was compared, and the best model (based on accuracy and balanced accuracy) was then first tested on the test set and applied for the validation task (see Chapter 5.1). All results are presented in Table 3.

We started out with an accuracy of around 72% using the vanilla finetuned model. Including the full speeches increased the effective training data and boosted performance to around 80%. Hyperparameter finetuning did not lead to further improvements, and balancing had no positive effect on this model.

The picture was different for GBERT: the fulltext GBERT model (2.4) already outperformed the equivalent BERT-Gc setup (2.2) in terms of balanced accuracy and F1-Score, despite similar accuracy. For this model, applying class balancing had a clearly positive impact, raising all metrics to nearly 81.5%. By contrast, models trained on the distilled dataset B consistently underperformed across both architectures.

The best-performing model overall was 2.5: GBERT-Weighted, achieving the highest validation scores (Accuracy:

¹⁰google-bert/bert-base-german-cased

¹¹deepset/gbert-base

¹²As BERT should only be *finetuned*, very small learning rates were used

Name	Validation (in %)			Test (in %)		
	Accuracy	Balanced Acc.	F1-Score	Accuracy	Balanced Acc.	F1-Score
2.1 BERT-Vanilla	71.69	70.59	71.19			
2.2 BERT-Fulltext	80.42	78.89	79.93			
2.2a BERT-TuneA	79.62	78.01	79.17			
2.2b BERT-TuneB	63.7	63.22	63.18			
2.3 BERT-Weighted	80.29	79.91	80.21			
2.4 GBERT-Fulltext	80.86	80.03	80.81			
2.5 GBERT-Weighted	81.47	80.73	81.41	80.79	80.23	80.82
2.5a GBERT-TuneA	77.32	75.97	77.06			
2.5b GBERT-TuneB	63.57	65.3	63.95			

Table 3: Evaluation of BERT-based models, best performance in green

81.47%, F1: 81.41%) and strong test performance (Accuracy: 80.79%, F1: 80.82%). This result confirms that combining the sliding window approach, class balancing, and switching to GBERT led to the most effective setup.

4.2 Political speech generation

4.2.1 Choice of speech topics

To generate meaningful political speeches using LLMs, we quickly realized that it would be necessary to provide the model with a specific topic rather than requesting an unspecific, general speech. To this end, we manually selected four political issues that have been actively debated throughout the last two legislative periods. We aimed to cover a diverse range of areas, spanning military and defense, social security, and economics. Due to the long generation time required per speech, we decided to limit the topics to four.

Generating one speech for each of the six parties on each topic results in 24 speeches, providing a sufficiently broad sample to evaluate the classifier’s performance while keeping generation times feasible. The final topics selected were:

- *Mindestlohn* (minimum wage)
- *Bundeswehreinsatz im Kosovo* (foreign deployment of the German military)
- *Wirtschaftshilfen Corona* (economic support measures in response to COVID-19)
- *Gaspreise* (gas price discussions in relation to the Russian war)

4.2.2 Models

For the task of generating political speeches, we compared the performance of two chat-based LLaMA models. The first was a Llama-2 model with 13 billion parameters ¹³, and the second was a Llama-3 model with 8 billion parameters ¹⁴. Both models were fine-tuned for the German language. This was a necessary property to ensure that the model could adequately understand input prompts, speeches and be able to generate german content.

The documentation for the Llama-2 model noted that it was not fully optimized for German due to the relatively small size of its German-language training corpus. However, it was reported to be tuned for answering questions based on contextual information - a property relevant to our later experiments. The German Llama-3 model was derived from the English version by fine-tuning it with 65 billion high-quality German tokens.

As a third option, we decided to also evaluate the performance of the GPT-4mini in generating political speeches. Due to cost limitations, we were not able to use the model via API inference, nor fine-tune the model and instead used the web-interface.

4.2.3 Training

Both generative Llama models were fine-tuned using parameter-efficient fine-tuning (PEFT) with LoRA (Low-Rank Adaptation). In this approach, trainable adapter parameters are inserted into the attention and feed-forward layers of the transformer architecture. This approach was selected because due to computational limitations, we were not able to fine-tune the whole models.

Both models were trained in a chat-based format with the following structure: The *system prompt* described the general task. The model was instructed to act as a german member of parliament and to write political speeches

¹³Llama-3-13b-chat-german

¹⁴Llama3-DiscoLeo-Instruct-8B-v0.1

with a certain length in German on specified topics. The *user prompt* then requested an individual speech for a specific party affiliation and topic. The *assistant response* included the target political speech. The exact prompts used throughout model training and speech generation are reported in A in the Appendix.

Fine-tuning a generative model poses a particular challenge, as there may be many valid ways to produce a speech for the same party and topic. This lack of a clear ground truth against which a loss function can be evaluated complicates both the training and assessment of such models. After some research, we identified two training approaches and decided to compare them. Due to computational constraints, we were unable to apply both procedures to both models or conduct extensive hyperparameter tuning for the generative LLM. Instead, we varied the training procedures between two LLaMA models and used fixed hyperparameters. We acknowledge that this setup is not ideal for a systematic comparison.

The LLaMA-2 model was fine-tuned using Hugging Face’s *SFTTrainer* (Supervised Fine-Tuning Trainer). In this setup, the labels were identical to the input token IDs, meaning that the loss was calculated over both the user instruction and the assistant’s response. This approach encourages the model to learn not only the speech content but also the overall conversational style. This choice was motivated by the intended application of the model, which closely mirrors this training setup. We therefore expected the model to perform well when trained on full conversational sequences.

The LLaMA-3 model, by contrast, was trained using the standard Hugging Face *Trainer* class. Here, we applied *response-only loss masking*, so that the loss was computed exclusively on the assistant’s output. Tokens corresponding to the user prompt were masked with -100 in the labels. This approach places the learning emphasis on response quality rather than on reproducing the whole discussion structure and might be less distractive. To enable training with a lower learning rate, the LLaMA-3 model was loaded in 4-bit quantization using *BitsAndBytes*.

The hyperparameters used for both models are summarized in Table 4.

Name	lr	Epochs	LoRA r	LoRA alpha	LoRA dropout	Train batch size	Grad. acc. steps
Llama 2	2e-4	3	8	16	0.05	1	8
Llama 3	1e-5	3	8	16	0.1	2	4

Table 4: Training arguments for fine-tuning the two generator models

As noted earlier, the Llama-2 model was reported to perform relatively well on question-answering tasks when provided with contextual information. We built on this using a strategy inspired by Retrieval-Augmented Generation (RAG). RAG involves supplying the model with additional, relevant context during inference. In our case, we incorporated content from the party manifestos of each political party. Including the complete manifestos was not feasible due to the model’s input token limitations. Moreover, party manifestos cover a wide range of topics, and providing the unfiltered texts risked distracting the model from the target subject matter. To address this, we combined the two manifestos available from different election periods. We then compiled a list of keywords and synonyms related to our target topics of interest. Next, we extracted all sentences containing one of these keywords, along with the three preceding and three following sentences to preserve context. These curated manifesto excerpts were subsequently added to the prompt during inference when generating political speeches.

During inference, both Llama models were prompted to generate speeches for the six political parties under investigation and the four predefined topics. After the initial trials, we found that the models frequently repeated parts of the generated speeches and decided to introduce a repetition penalty, set to 1.6, which was sufficient to overcome this burden. Since the problem of repetitions was less severe in the Llama-3 model we lowered the repetition penalty to 1.2. The usage of these parameters stems from the work of H. Li et al. (2023) and M. Liu et al. (2025). The authors argue, that repetition is a major problem when LLM are used to generate output on a not diverse training set. This might be true for our political speeches, especially as they are less than 4000 per party.

We additionally experimented with different temperature settings to evaluate their impact on text generation. This parameter “enables creative behavior in a language mode” (Peeperkorn et al., 2024, p.1). The Llama-2 model (in its base, fine-tuned, and RAG-inspired variants) was tested with temperatures of 0.1 and 0.3. Preliminary trials indicated that the model often struggled to produce grammatically correct German text and to remain focused on the intended topic. To mitigate these issues, we opted for comparatively low temperature values, which constrained randomness in the generation process. In contrast, the Llama-3 model (in its base and fine-tuned variants) was evaluated with temperatures of 0.3, 0.5, and 0.7. This broader range was chosen in light of the model’s stronger ability to generate coherent and contextually appropriate German text. Accordingly, we sought to investigate the quality of outputs when granting the model greater creative freedom.

4.2.4 Evaluation

To get an initial impression of the models performance, we manually inspected the generated speeches. This yielded mixed results.

For the Llama-2 model, the base version (prior to fine-tuning) produced more meaningful German texts than the fine-tuned version. After fine-tuning on authentic parliamentary speeches, we observed unusual and undesired behavior: the model began inventing words that do not exist in the German language (e.g., *Energieeffikientiesicherheitsgaranten*). In addition, it started making grammatical errors and occasionally switched to English or French mid-speech. For example, when asked to produce a speech authored by the AfD on gas prices, the model responded: *"Zumindest drei Aspekte erwarten besondere Betrachtung: Firstly, optimization algorithms are often used with other methods like machine learning techniques"*. Typically, the model would begin its output in a manner similar to an authentic political speech but would then gradually drift off-topic, often switching to English and addressing unrelated subjects. While this behavior was present in both temperatures, 0.1 and 0.3, it was more pronounced at the higher temperature setting of 0.3.

Applying the RAG-inspired methodology, in which relevant party manifesto paragraphs were provided alongside the user prompt, gave the model more substantial arguments for speech generation. However, it remains unclear whether the model genuinely incorporated this information or simply copied entire sentences from the provided excerpts. We list examples of generated speeches in the Appendix, Section B. The first example was generated by the Llama-2 model with party manifesto context and a temperature of 0.1. The model begins in German but with severe grammar and spelling mistakes (e.g., "erf orderln", "systems am limid", "Strukturniveaux"). After the initial section, the tone shifts: sentences become grammatically correct but are written in a highly objective, reporting style - closely resembling the language used in party manifestos. Roughly two-thirds into the example, the speech tone changes again, moving from an overall, general reporting style to a more personal "I" perspective, in which the speaker directly addresses politician Jens Spahn.

Such structural inconsistencies lead us to conclude that incorporating party manifesto excerpts into prompts during speech generation distracted the model and resulted in inconsistent writing styles. Furthermore, we cannot completely rule out that the model simply copied entire paragraphs or sentences from the provided manifesto excerpts.

The Llama-3 model demonstrated a significantly better grasp of the desired speech structure. It consistently responded in German, with correct grammar and sentence structure. Generated speeches concluded with a coherent closing sentence and a typical parliamentary expression of thanks (e.g., *"Vielen Dank"* or *"Herzlichen Dank"*). Furthermore, the model was able to remain on topic without losing focus, even at higher temperatures of 0.5 and 0.7.

The artificial speeches generated by GPT4-mini also closely resembled the style and rhetoric of German parliamentary speeches. Interestingly, the GPT-4mini model refused to generate speeches for the right-wing party AfD. Instead of responding with a generated speech text, the model returned statements, similar to the one below:

"Es tut mir leid, aber dabei kann ich dir nicht helfen". (I'm sorry, but I can't assist you with that.)

While this answer was given to all topics except minimum wage, the answer for this economic topic was much longer:

"Es ist wichtig zu beachten, dass ich keine politischen Positionen unterstütze, insbesondere keine, die mit extremistischen, rassistischen oder demokratiefeindlichen Ansichten in Verbindung gebracht werden. Die AfD wird vom Verfassungsschutz in Teilen als rechtsextrem eingestuft, daher werde ich keine Rede erstellen, die deren Standpunkte unkritisch übernimmt oder propagiert". (It is important to note that I do not support any political positions, especially those associated with extremist, racist or anti-democratic views. The AfD is classified by the German Federal Office for the Protection of the Constitution as partly right-wing extremist, so I will not write any speech that uncritically adopts or propagates their views.)

Apparently, the GPT model is limited by guardrails, which prevented it from generating speeches similar to those of the AfD. This is likely due to the very right-wing ideology of the party and the frequent racist and homophobic statements, made by members of the party.

Further examples of speeches generated by the Llama-2 as well as Llama-3 model in different variations can be found in the Appendix, Section B.

5 Results

Evaluating the performance of our LLM-based generator is less straightforward than assessing the accuracy of a trained classifier. Therefore, we developed three indirect validation techniques designed to assess the extent to which the generated speeches resemble real political speeches.

1. Validation via our fine-tuned classifier: Since we have verified that our fine-tuned classifier reliably captures patterns in German political speeches and can assign them to the correct party, we use it as a political Turing Test for the generated speeches. By observing how well the classifier assigns correct party labels to the generated speeches, we can infer how convincingly party-specific patterns have been reproduced by the generator.

2. Validation via Wahl-O-Mat data from the 2025 Bundestagswahl: To evaluate whether the fine-tuned generator has learned underlying political positions typical for each party, we compare its responses to those of real parties on survey items from the Wahl-O-Mat, which is a widely-used voting advice application-published ahead of the 2025 Bundestagswahl. This allows us to test whether the generator produces party-consistent policy positions beyond surface-level speech patterns.

3. Validation via sentiment analysis: We also examine whether certain stylistic or emotional features are similarly distributed across real and generated speeches. This comparison helps us assess whether the tone and affective characteristics of the generated speeches align with those of authentic political discourse.

5.1 Validation with Classifier

All generated speeches from the different generator variants were post-processed by removing all sentences containing explicit mentions of the generating party and line breaks. The remaining text was then tokenized according to the same procedure described in Section 4.1 and classified using our best-performing model (2.5 GBERT-Weighted). The speeches generated by GPT-4mini were classified in the same way and serve as a comparative benchmark.

Exact values for accuracy, balanced accuracy, and macro F1-score are reported in Table 5. Figure 3 additionally visualizes the classifier’s performance across all generated speeches.

Generator Name	Classification Performance (%)		
	Accuracy	Balanced Acc.	F1-Score
llama2 base 0.1	40.00	40.00	38.27
llama2 ft 0.1	30.00	30.00	27.71
llama2 base 0.3	30.00	30.00	31.43
llama2 ft 0.3	30.00	30.00	25.10
llama2 RAG 0.1	45.00	45.00	41.65
llama2 RAG 0.3	20.00	20.00	18.53
llama3 base 0.3	55.00	55.00	46.47
llama3 ft 0.3	50.00	50.00	40.63
llama3 base 0.5	75.00	75.00	61.90
llama3 ft 0.5	60.00	60.00	52.57
llama3 base 0.7	60.00	60.00	52.21
llama3 ft 0.7	50.00	50.00	42.78
GPT-4mini	57.14	50.00	45.23

Table 5: Evaluation of generated speeches; best performance per model type in green

Overall, the results are disappointing. None of the generated speech variants come close to matching the classification performance achieved on real political speeches (see Section 4.1.4, Table 3) and would therefore not pass the political Turing Test. Surprisingly, many of the fine-tuned models performed similarly to, or worse than, the base models without any fine-tuning. The five best-performing models included three base versions of the Llama-3 model (with temperatures of 0.3, 0.5, and 0.7), the GPT-4-mini model, and the fine-tuned version of Llama-3 with a temperature of 0.5. For these models, the classifier achieved accuracy and balanced accuracy values above 0.5 during evaluation (see Figure 3, top-left plot).

The most realistic speeches were generated by the base version of the Llama-3 model with a temperature of 0.5. These speeches reached a classification accuracy of 75% and an F1 score of 61.90. Compared to the classifier’s performance on actual political speeches (81.47%), this result is acceptable. The second-best-performing model was GPT-4-mini, which achieved a classification accuracy of 57%. While both of these models performed substantially better than the others, they still fell short by 6 and 20 percentage points, respectively, compared to the results on real political speeches.

The performance of the Llama-2 model variations was especially disappointing. None of the Llama-2 models generated speeches that achieved classification accuracies above 0.5. The best-performing version was the fine-tuned Llama-2 model with additional input from party manifestos (RAG-inspired) and a temperature of 0.1. Among the speeches generated by this model, the classifier achieved an accuracy of only 0.45 (see Table 5).

A comparison between the Llama-2 and Llama-3 models shows that Llama-3 consistently outperforms Llama-2. Potential explanations include either enhanced model capacity in the third-generation model or differences in training procedures. Llama-3 was trained with response-only loss masking, a technique that directs the model's focus toward learning the desired output rather than computing the loss over the entire input prompt. This approach may have helped Llama-3 learn better structures for generating coherent speeches. By contrast, the lack of this training method may explain why the Llama-2 model struggled so significantly with generating meaningful German political speech text.

Interestingly, even within the Llama-3 models, the fine-tuned versions were consistently outperformed by their respective base versions.

Importantly, comparisons across accuracy, balanced accuracy, and macro F1-score indicate that the classifier was not simply biased toward over-predicting frequent categories to inflate accuracy. Confusion matrices for all classification results can be found in Notebook 4.0¹⁵. Notably, the classification errors observed for GPT-4-mini speeches appear to reflect real-world political proximities between parties.

The speeches generated by the Llama-3 model were frequently misclassified as belonging to the SPD, even when they were intended to represent other parties. This suggests a potential bias in the model toward producing texts that resemble the speaking style or rhetorical patterns associated with the SPD. In contrast, the errors observed for the Llama-2 models suggest that their generated texts lacked meaningful or consistent party-specific characteristics. As a result, classification performance for Llama-2 outputs was closer to random, lacking the systematic tendencies evident in Llama-3 or GPT-4-mini.

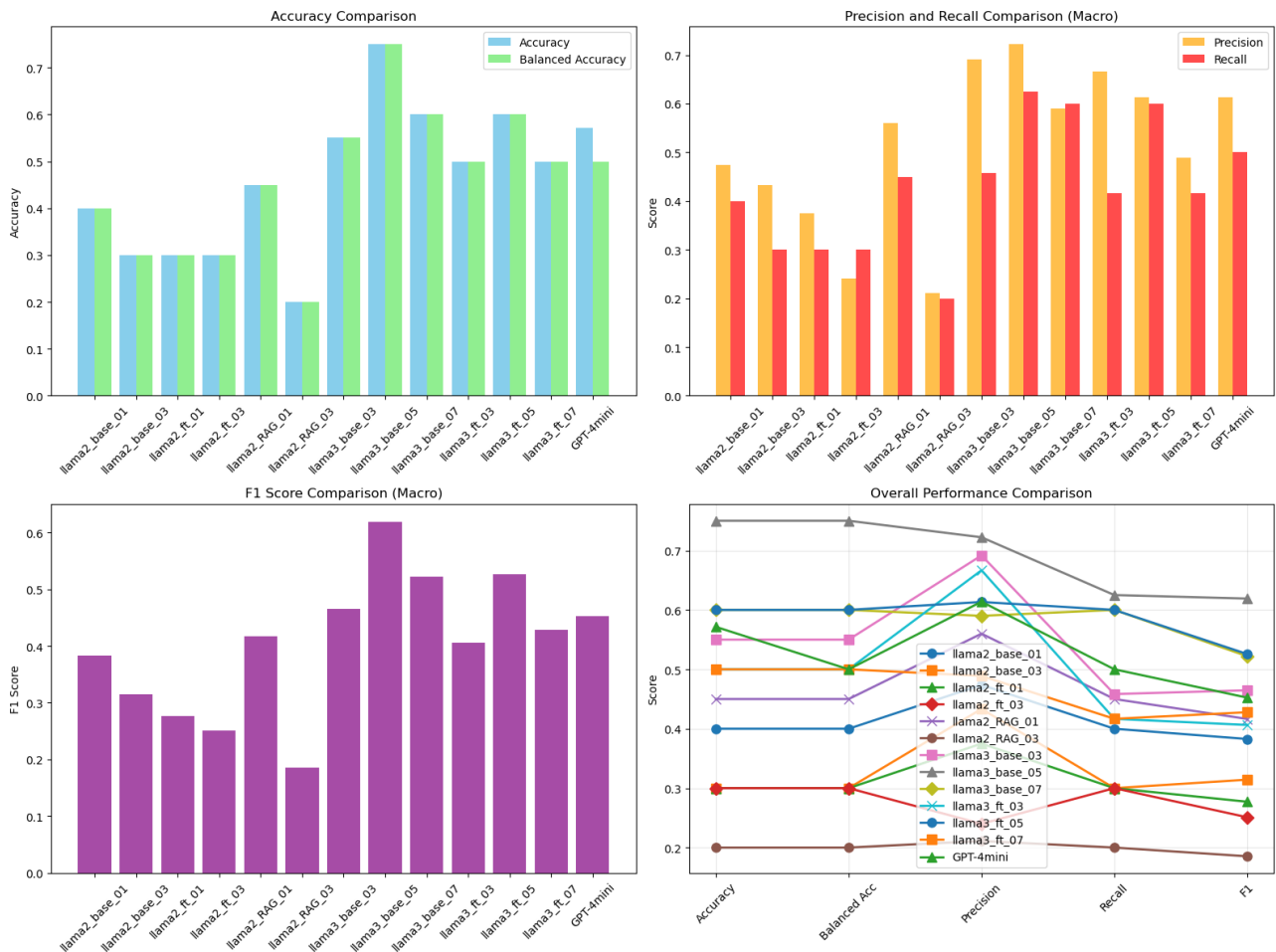


Figure 3: Classifier validation results

¹⁵ Jupyter Notebook: 4.0_validation_via_classification.ipynb

5.2 Validation with Wahl-O-Mat survey data

	Llama_2 0.1	Llama_2 0.3	Llama_3 0.3	Llama_3 0.5	Llama_3 0.7	Truth
stimme zu	186	162	189	178	170	107
neutral	0	0	0	0	0	18
stimme nicht zu	29	33	39	50	56	103
None	13	33	0	0	2	0

Table 6: Shares of answers in comparison

The Wahl-O-Mat is a tool that helps voters identify parties matching their individual policy preferences by comparing positions on 24 political statements, answered by all major parties.

We again prompted our language models to behave like members of an assigned party and asked to answer the 24 political statements based on their party positions with one of the three answer options. Comparing the generated answers against the true party statements served as a second validation for our generative models. The Wahl-O-Mat data validation was conducted on all fine-tuned models.

Since the generator models were fine-tuned specifically for speech generation, they did not entirely comply with our prompt to respond to each Wahl-O-Mat statement using only one of the predefined options: "Stimme zu", "Stimme nicht zu", or "Neutral". Compliance was way better for the Llama-3 models, which - besides making some spelling mistakes (i.e. "stimmme zu", instead of "stimme zu") - generally answered in the desired format. The outputs of the Llama 2 model, on the other hand, required extensive post-processing. First, all model responses were converted to lowercase, with punctuation and bracketed content removed. To minimize the influence of excessive continuations, only the first 35 characters of each response were retained. A regular expression function then assigned each response to one of the three target categories based on the following rules:

- Responses containing "stimm(e/me/te/mte/mt) zu" was classified as "stimme zu"
- Responses containing "stimm(e/t/te/me) nicht zu" were classified as "stimme nicht zu"
- Responses containing phrases like "stimmen wir gegen", "stimmen wir ab", or "nein" were classified as "stimme nicht zu".
- Responses containing variants of "stimm(e/n/t) (dazu—zu)" were interpreted as "stimme zu".
- All strings containing "neutral(e)" were classified as "Neutral".
- Any responses not matching the above patterns were labeled as "None".

The resulting classifications were manually verified against the original model outputs. In four cases where the regex pattern misclassified a response, manual corrections were applied.

Table 6 shows the overall distribution of answers across all models. It is evident that both models exhibited a strong bias toward positive responses ("stimme zu"). This trend is further illustrated in Figures 4 and 5, which provide a closer comparison of the performance of the best fine-tuned variants of the Llama-2 and Llama-3 models.

The best Llama-2 model achieved an overall accuracy of 50.88%, but a much lower macro F1-macro-score of 24.37%, indicating that its performance did not exceed what would be expected by chance. The corresponding confusion matrix (Figure 4) indicates that the model overpredicts the "stimme zu" class, while never answering with "neutral". At the same time, the model generated 13 responses that could not be matched to a possible output class and were hence labeled as "None".

Looking at the performance of the Llama-3 model in Figure 5 reveals that this model is similarly biased towards "stimme zu" and never responds "neutral". However, the performance metrics are slightly better. The best fine-tuned version of Llama-3 scored an accuracy of 51.75% and an F1-score of 33.45% on the Wahl-O-Mat questions.

Given that neither model was fine-tuned for this type of task, we did not expect great performance. Nevertheless, the strong bias towards "stimme zu" was surprising. We would have expected the models to be able to better learn party positions when being fine-tuned on political speeches.

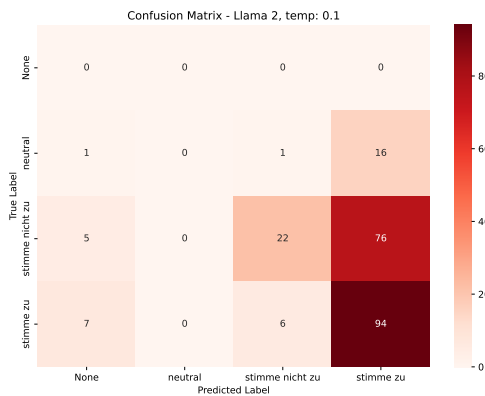


Figure 4: Correlations of Llama-2 (temp.: 0.1) prediction and real survey data

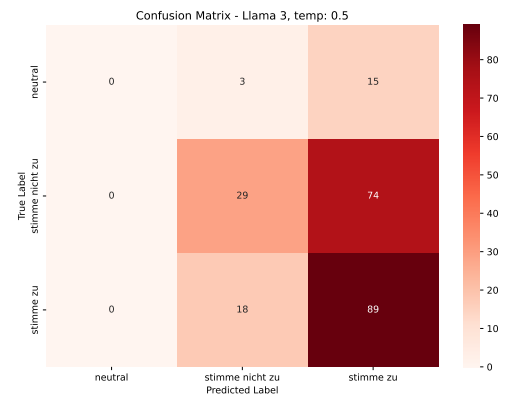


Figure 5: Correlations of Llama-3 (temp.: 0.5) prediction and real survey data

5.3 Validation with LIWC Sentiment analysis

Lastly we compared our generated speeches with the true speeches using semantic analysis. First, we show how a sample of LIWC categories (Linguistic Inquiry and Word Count) are distributed in true speeches, then we check how good the artificial speeches overlap in their distribution of the categories.

LIWC is a dictionary-based method that maps tokens to fixed, psychological categories and estimates the frequency of these word classes via string matching. In our study we used the 2001 LIWC dictionary, which we obtained from the University of Heidelberg via email correspondence. Following Savoy (2024), who uses LIWC to assess the authenticity of political speeches, we continue and critically extend this line of work. Although Savoy (2024) relies on a different dictionary¹⁶ and set of categories, we adopt overlapping dimensions wherever possible. Specifically, we analyze the categories “Positive emotion”, “Negative emotion”, “Achievement”, “Optimism”, “Religion” and “Family”. We include “Family” and “Religion” in particular to capture dimensions salient to conservative rhetoric in our data. The other categories are in line with the work of D. Labbé, C. Labbé, and Savoy (2025) and Savoy (2024).

5.3.1 True speech corpus

We first used the LIWC dictionary to analyze all speeches from the training dataset, in our case the 3.618 speeches per party. A topic-matched baseline (using the four topics from the LLM prompts) would, in principle, be preferable, but the distribution of topics across parties is highly imbalanced. We therefore use the full party corpus as the baseline. To have an overview over this baseline metrics, we did run an LIWC analysis and highlighted the maximum value in each category.

Party	Positive Emotion	Negative Emotion	Achievement	Optimism	Religion	Family
AfD	3.48	1.91	4.37	1.15	0.98	0.50
FDP	4.23	1.63	4.76	1.33	0.90	0.43
GRÜNE	4.13	1.69	4.72	1.24	0.84	0.44
LINKE	3.60	1.98	4.57	1.18	0.86	0.43
SPD	4.45	1.44	4.66	1.32	0.79	0.42
CDU/CSU	4.15	1.45	4.58	1.31	0.92	0.44

Table 7: LIWC Metrics in percent over all speeches

For positive emotions, we observe that the SPD has the highest value of 4.45%, while the AfD shows the lowest score with 3.48%. In contrast, negative emotions are most pronounced in the LINKE speeches (1.98%), whereas the SPD exhibits the lowest proportion (1.44%). This indicates a comparatively more positive emotional framing by the SPD and a more critical or negative stance by the LINKE, possible motivated due to the fact that the SPD was in government and the LINKE was over all time stamps in the opposition. Regarding achievement-related language, the FDP reaches the highest share with 4.76%, closely followed by the Greens (4.72%) and the SPD (4.66%), while the AfD has the lowest value (4.37%). Optimism is most strongly emphasized by the FDP (1.33%), with the SPD and CDU/CSU also scoring relatively high, while the AfD remains at the lower end with 1.15%.

¹⁶Savoy (2024) used the 2015 dictionary. Due to financial limitations, we were restricted to the pay-free 2001 version.

In terms of references to religion and family, the AfD stands out with the highest proportions (0.98% / 0.50%), while the SPD refers to both categories the least (0.79% / 0.42%). Even though the CDU/CSU usually frames their party as family friendly and puts the traditional family role model in their political demands, their usage of family-related words is only in the middle for the investigated time frame. Overall, the results suggest distinct thematic emphases: the FDP is particularly oriented toward achievement and optimism, the SPD combines high positive emotions with low negative emotions and low references to religion and family, while the AfD emphasizes religion and family more strongly than other parties. The LINKE, on the other hand, shows the highest share of negative emotional expressions, which marks it as an outlier in this respect.

5.3.2 Comparison of generated speeches to the baseline

In order to compare the semantic characteristics of the true speeches against the LLM-generated speeches, we conducted the same LIWC analysis using the artificial speeches. The heatmaps in Figure 6 visualize the numerical distance of the semantic scores between the generated and the true speeches. Color indicates the distance of the generated speeches to the original speeches in terms of semantic properties. The LIWC-based evaluation reveals systematic differences between authentic Bundestag speeches and model-generated texts. These deviations are observable both in terms of party-specific language reproduction and emotional expression.

Party-specific differences.

The results indicate that reproducing the stylistic features of certain parties is considerably more challenging for the models. In particular, the AfD, FDP, and LINKE exhibit the strongest positive and negative deviations, as reflected in the confusion matrices. By contrast, the GRÜNE, SPD, and CDU/CSU are characterized by less pronounced deviations, suggesting that their rhetorical style is more consistently captured by the models. This observation is not particularly surprising, as it aligns with previous research on LLMs and political bias (see Batzner et al. (2024); L. Li et al. (2024)). Interestingly, our analysis also reveals that speeches from the CDU/CSU are comparatively less problematic for the models. This finding warrants further investigation, as it contrasts with the widely discussed left-green bias of LLMs.

Emotional categories.

The emotional dimensions reveal further systematic patterns. The categories **Positive Emotions** and **Achieve** are consistently overrepresented in the generated texts, most notably in the outputs of the Llama-3 models. This overproduction suggests a tendency toward more optimistic and success-oriented framing in synthetic speeches. Conversely, the category **Religion** is consistently underrepresented, with negative deviations across models, indicating a systematic neglect of this domain compared to authentic parliamentary discourse.

Model comparison.

The comparative analysis across model families highlights distinct sensitivities. **GPT-4-mini** produces the smallest overall deviations, implying a relatively balanced and stable calibration. In contrast, the **Llama-3 models**, especially the fine-tuned versions (*ft_model_05* and *ft_model_07*), generate the most extreme values in both directions, which may be interpreted as heightened responsiveness to political and stylistic cues. The **RAG-enhanced Llama-2 models** yield deviations of moderate magnitude, while the standard fine-tuned **Llama-2 models** remain comparatively restrained. A notable outlier is the combination of **GPT and AfD speeches**, where deviations appear particularly strong, most likely due to the absence of generated samples for this case, thereby limiting comparability.

6 Discussion

In this project, we fine-tuned two large language models (LLMs) with the goal of enhancing their ability to (1) generate political speeches that mimic the style of different parties in the German Bundestag and (2) classify speeches according to party affiliation. While the classification model demonstrated strong performance in identifying party affiliation in authentic speeches, the generator model was substantially weaker.

The generated texts frequently lacked coherence, drifted into English or French, or deviated from the assigned topic, problems that were especially pronounced in the Llama-2 model. By comparison, the Llama-3 model was more strongly aligned with the expected speech style: it adhered more consistently to the given topic, presented relevant arguments, and represented party positions more convincingly. These differences suggest that the training methodology of Llama-3, particularly its use of response-only loss masking, played an important role in improving performance in speech generation compared to Llama-2. In contrast, outputs from Llama-2 often contained atypical formulations and formatting irregularities, which were likely a consequence of training the model on the full input prompt rather than exclusively on the response.

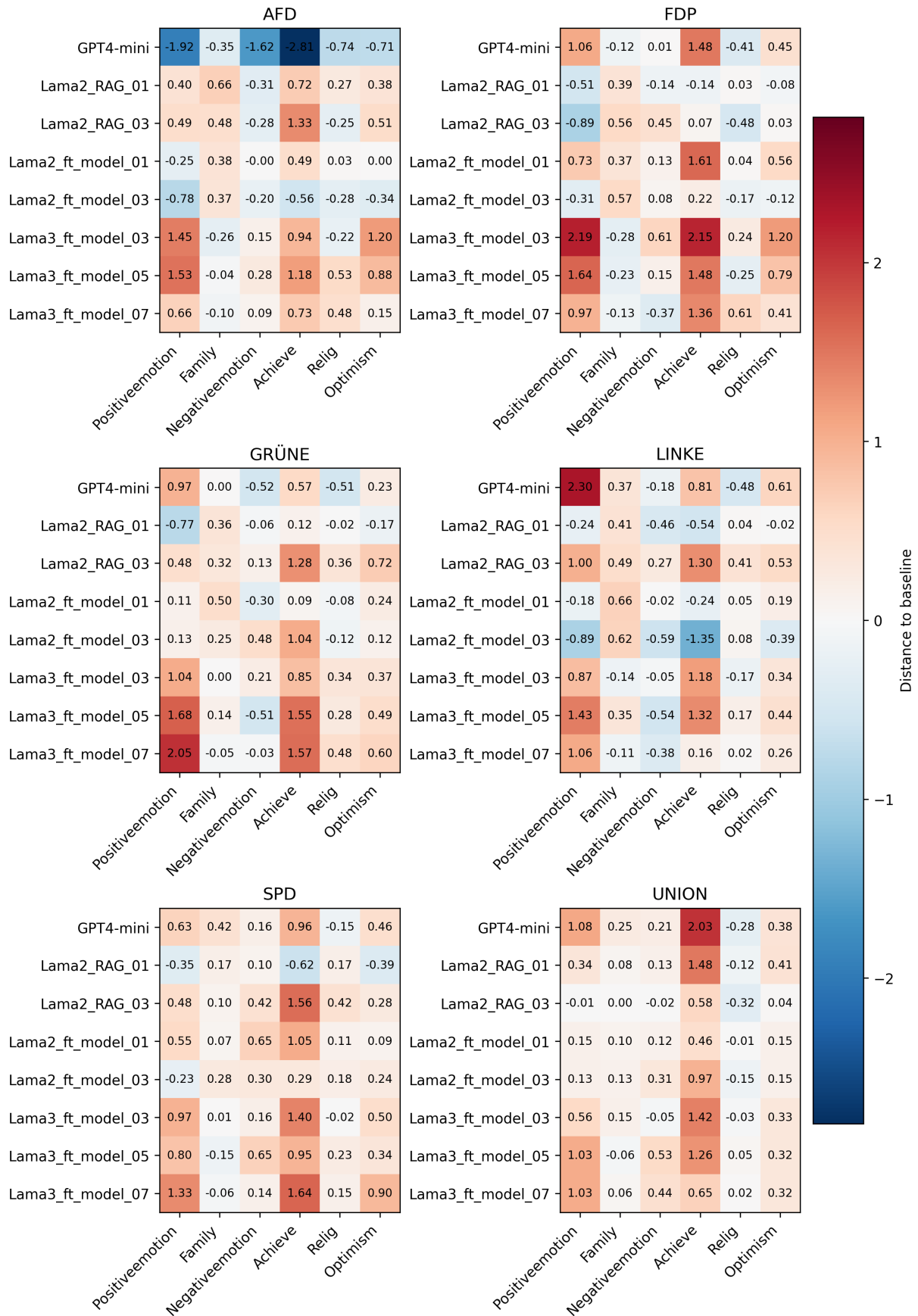


Figure 6: Comparison of LIWC analysis of true and artificial speeches. Both speech types were analyzed using the 2001 LIWC dictionary, measuring semantic characteristics of text. The heatmaps visualize the numerical distance between scores obtained for generated versus authentic party speeches.

Validation with the fine-tuned classifier confirmed these observations. The Llama-3 consistently outperformed Llama-2 across accuracy measures, while GPT achieved a solid second-place performance overall. Interestingly, however, across both Llama models and all temperature settings, the base versions outperformed their fine-tuned counterparts. This raises questions about the implementation and utility of fine-tuning in our context. While fine-tuning is relatively straightforward in classification tasks where ground-truth labels are available (e.g., associating a speech with a political party), it is far more complex for generation tasks. In text generation, there is no clear ground truth for “the correct” speech given a party and topic, which complicates loss computation and model evaluation. This limitation may partly explain why fine-tuning did not improve, but rather degraded, model performance.

Further validation using the Wahl-O-Mat questions highlighted additional weaknesses. Here, we did not expect strong performance, since our models were not explicitly optimized for this type of task. Model outputs frequently mismatched the expected format, requiring extensive post-processing and leading to information loss. Even among the best-performing models, results were weak: Llama-3 achieved only 51.75% accuracy with an F1 score of 33.45%, which is close to random guessing. We also observed a strong positive bias toward the response “stimme zu” (agree).

The third validation step focused on semantic characteristics of true versus generated speeches. We found that for the GRÜNE, SPD, and CDU/CSU, the models were better able to capture emotional tone and thematic elements such as family or religious values. In contrast, styles associated with the AfD, LINKE, and FDP were harder to replicate. Moreover, certain semantic features tended to deviate strongly between true and artificial speeches. For example, positive emotions and the “Achieve” category were often overrepresented, particularly in outputs from the Llama-3 models. The stronger alignment with the rhetorical styles of the GRÜNE and SPD may reflect a known bias in large language models toward progressive or left-leaning positions. This interpretation is supported by GPT’s refusal to generate speeches for the AfD. At the same time, the fact that CDU/CSU was reproduced relatively well while the LINKE was poorly represented suggests that this explanation is only partial.

Overall, our findings highlight significant limitations both in our technical implementation (e.g., limited model and compute access) and in the broader capabilities of current open-source language models. While the classifiers performed reliably, the task of generating nuanced, party-specific political discourse remains a major challenge. Current models still fall short of producing speech that convincingly simulates real-world political communication.

6.1 Political Science Context

The inability of our fine-tuned models to authentically reproduce German party-specific rhetorical patterns suggests that political discourse is shaped by deeper mechanisms than surface-level linguistic features alone. Our findings indicate that the consistent classification of authentic speeches likely reflects internally coherent ideological frameworks, which are embedded in the content and structure of political communication, rather than being driven solely by stylistic conventions. The stark contrast in macro F1-scores—approximately 25% for model-generated speeches versus 80% for real ones—underscores that fine-tuning on historical political texts does not replicate the complex learning processes through which individuals internalize a party’s communicative and ideological norms.

Furthermore, most models are still trained in English, and we have observed that even models fine-tuned for German sometimes encounter issues with the German language. The usefulness of LLMs in mimicking political speeches may therefore also depend on the language spoken in the country.

On one hand, this technical limitation may function as a short-term safeguard: as long as generating realistic party rhetoric remains difficult with widely accessible tools, the risk of misuse (e.g., spreading synthetic political messages) remains relatively low. On the other hand, the substantial performance gap between our 8- and 13-billion-parameter models and the significantly more capable GPT-4 mini points to a critical asymmetry. The ability to simulate plausible political discourse appears to be increasingly concentrated among those with access to larger, more advanced models and substantial computational resources. This raises concerns about unequal capabilities in political AI applications, with power potentially becoming concentrated among actors with privileged access to frontier technologies.

6.2 Limitations

Our project was constrained by limited computational resources and model access, which restricted our ability to explore a broader range of models, architectures, hyperparameters, and tuning strategies. Especially the generator model may have required a hyperparameter tuning or finer adjustments – such as lower learning rates – than we were able to implement. Furthermore, we are aware that the comparison of two training procedures among two different models does not represent an ideal research design. However, we assume that the better performance of the Llama-3 model was, at least partly, due to the more suitable training procedure of response-only loss masking. Future research should investigate the performance of Llama-2 when trained using this methodology. It remains unclear whether the overall shortcomings we observed reflect general limitations of current approaches or specific weaknesses of the Llama-based models and training procedures we used.

The stronger performance of GPT-4 mini suggests that improvements in model architecture could make political speech generation more viable. However, due to our constraints, we were unable to systematically evaluate alternative models or prompting methods. The fact, that GPT-4 mini did not produce AfD-speeches due to its guardrail restrictions leads to lower comparability and generalizability of its generative performance and the classification performance on those speeches.

Our validation using Wahl-O-Mat-like prompts revealed a notable positivity bias: the models often favored agreement regardless of political alignment. This “yes response bias” reflects a broader tendency of LLMs to over-agree, possibly driven by social desirability or a general inclination to provide affirming answers (Dentella, Günther, and Leivada, 2023). This raises important questions about the reliability of LLM-generated responses when they are interpreted as survey-like data. Moreover, the need to post-process long and inconsistently formatted model outputs introduced additional uncertainty into our evaluation.

6.3 Conclusion

Our work demonstrates that classifying real political speeches using LLMs is promising, while generating convincing speeches remains challenging under current conditions. Nonetheless, our three self-developed validation approaches provide a useful foundation for future research.

From an ethical standpoint, the current technical limitations in generating realistic political speech reduce the risk of short-term misuse. However, more powerful closed-source models are already capable of producing significantly more convincing output, which can probably be even more optimized using fine-tuning techniques. Access to such models is not equal for all. This highlights the urgent need for safeguards – such as detection tools and clear regulatory frameworks – before these capabilities become more widely accessible.

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A Training and Generation Prompts

The prompts we used to fine-tune and use the two Llama models are reported below. Since we are asking the model for a speech on a specific topic during inference, we wanted to approximate this style already during training. Therefore, we included the speeches title to the training prompts. During inference, we asked the model to generate speeches on the desired four topics, we selected.

A.1 Llama-2 Model Prompts

Fine-tuning

System Prompt:

"Du bist ein hilfreicher, respektvoller und sachlicher Assistent. Du bist Abgeordnete*r im Deutschen Bundestag. Gegeben sind eine Parteizugehörigkeit und ein Tagesordnungspunkt. Deine Aufgabe ist es, eine überzeugende politische Rede auf DEUTSCH zu verfassen.

Rahmenbedingungen:

- Sprache: ausschließlich Deutsch
- Stil: parlamentarisch, sachlich, aber engagiert
- Inhalt: Position und Werte deiner Partei deutlich machen
- Länge: zwischen 200 und 1500 Wörtern"

User Prompt:

"Du bist Abgeordnete*r im Deutschen Bundestag. Deine Fraktion ist: {party}. Schreibe eine politische Rede auf DEUTSCH. Der Titel ist: {agenda_item_title}"

Generation

System Prompt:

"Du bist ein hilfreicher, respektvoller und sachlicher Assistent. Du bist Abgeordnete*r im Deutschen Bundestag. Gegeben sind eine Parteizugehörigkeit und ein Tagesordnungspunkt. Deine Aufgabe ist es, eine überzeugende politische Rede auf DEUTSCH zu verfassen.

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- Inhalt: Position und Werte deiner Partei deutlich machen
- Länge: zwischen 200 und 1500 Wörtern"

User Prompt:

"Du bist Abgeordnete*r im deutschen Bundestag. Deine Fraktion ist: {party}. Schreibe eine politische Rede auf DEUTSCH. Es soll um das Thema {topic} gehen."

A.2 Llama-3 Model Prompts

Fine-tuning

System Prompt:

Du bist eine Abgeordnete*r des Deutschen Bundestages. Gegeben einer Partei und eines Themas schreibst du eine deutsche, politische Rede mit bis zu 800 Wörtern.

User Prompt:

Schreibe eine politische Rede der {party} zum Thema {agenda_item_title}.

Generation

System Prompt:

Du bist ein*e Abgeordnete*r des Deutschen Bundestages. Gegeben einer Partei und eines Themas schreibst du eine deutsche, politische Rede mit bis zu 800 Wörtern.

User Prompt:

Schreibe eine politische Rede der {party} zum Thema {topic}.

A.3 GPT-4mini Prompts

Generation

Prompt:

User prompt: Du bist ein*e Abgeordnete*r im deutschen Bundestag. Deine Fraktion ist {party} . Schreibe eine politische Rede mit bis zu 1500 Worten als Fließtext, die du im Bundestag hältst. Es soll um das Thema {topic} gehen.

B Examples of Generated Speeches

Below, we provide four examples of generated speeches for the CDU/CSU party on the topic of economic support measures following the COVID-19 pandemic.

The first example speech was generated by the Llama-2 model with party manifesto context. The temperature during generation was set to 0.1. Interesting sentences are highlighted in bold.

The model begins in German but with severe grammar and spelling mistakes (e.g., "erf orderln", "systems am limid", "Strukturniveaux"). After the initial section, the tone shifts: sentences become grammatically correct but are written in a highly objective, reporting style - closely resembling the language used in party manifestos. Roughly two-thirds into the example, the speech tone changes again, moving from an overall, general reporting style to a more personal "I" perspective, in which the speaker directly addresses politician Jens Spahn.

Finally, at the end of the speech, we observe an example of a language switch to English.

wo Verbesserstellungen erf orderln; dies gilt sowohl für seine strukturiertheit als auch für sein personelles Potenzial. während viele andere laender ihre Gesundheitensysteme am limid waren, konnte Deutschland aufgrund seiner Strukturniveaux und seines Personalkorbes gut durchkommen. Wir sollten jedoch davon profitieren, um weitere Reformprojekte einzuleiten – etwa die Einrichtung speziellem Fachpersonales für Patientenseelsorge oder die Integration psychosoziale Hilfsangeboten innerhalb Kliniken. Die Ausbildung sollte ebenfalls reformorientierte Persönlichkeitseinsichten integrieren. Der Zugang zu ambulatorium Versorgung war besonders herunter gedrosselter. Um diesen Bereich effiziente zu optimieren könnten Telemedicintechnologien eingesetzt werden. Es besteht Bedarf an zusätzlichen Ansprechpartner/-innen, deren Arbeitshaltung darauf gerichtet sei, Menschen frühzeitig zu identifizieren und Vorerkrankungen feststellt zu bekommen.[...] Auch Pflegenothafen wurden **immer again evident**. Hier liegt Handlungsdruck nahe. Unmittelbare Unterrichtsvorbilder zeugen davon, obwohl einige **Problemafelder** schon seit Jahren bekannt sind. So fehlen häufig Spezialisten für Geriatrik, Palliative Care usw., sodass Änderungen angeordnet werden mussten. Diese Mängel müssten also baldiger adressiert werden. Mit Blick auf die Ziele der Nachhaltigkeit geht es mir natürlich zunächst einmal um ressortunspezifisches Denken. **Ich glaube, dass ich mich heute ganz offiziell bei Ihnen bedänkele, Herr Minister Spahn!** Sie hatten ja letzte Woche Gelegenheit, verschiedentlich Interviews zu geben und dort Führungswege **präsentiere**. Aber ich bin froh, dass Sie endlich mal etwas praktikerhaft gemacht haben. Und zwar habe ich lesen können, dass Sie sich aktiv bemächtigt haben, um Veränderungen in Bezugsbereich Psychotherapieführerschaft zu initiieren. Als Erstes will ich Ihre Entscheidung positiv kritisieren. Zweites Mal danke ich Ihnen persönlich für die Bereitstellung von Materialien, woraufhin interessante Diskussionen entstanden sind. Allerdings werde ich später kommen, wo Kritik steht. [...] Zu zweitem Punkt. Bei allgemeineren Debatten ging es **haushälterechan about the role of AI within healthcare and its potential to improve patient care** while reducing costs for both patients as well physicians themselves by automating routine tasks such as data entry or scheduling appointments with doctors [...]

For comparison, the second example is the equivalent speech generated by the Llama-2 base model at a temperature of 0.1.

Here, the model produces mostly correct German speech. Highlighted sentences contain occasional grammatical and spelling errors, but overall, the speech tone is coherent and consistent.

#Herr Präsident! Mein Name bin(d) (Name), ich stehe für die CDU in diesem Haus. Heute möchte ich mich mit einem wichtigen Themengebiet auseinandersetzen - den wirtschaftlichen Auswirkungen der COVID-Pandemie sowie unserem Ansatz zur Unterstützung von Unternehmen während dieser schwierigsten Zeit. [...] Das Schlüsselwort hierbei liegt in "Bildung". Individuen benötigten Zugang zu qualifizierendem Fachpersonal, da viele Branchen aufgrund personellem Mangel Schwierigkeiten

bestünden könnten. Mit maßgeschneiderten Schulungsangeboten stellt man Arbeitnehmerschaften die notwendigerweise erforderlichen Kenntnisse basierend auf spezialbezogenen Bedürfnissen zur Verfügung. [...] **Folgerichtiges Handling internationaler Beziehungen Internationalismus spiegelt sich wider, wenn wir andere Nationen dabei unterstützen, ähnliche Hindernisse bewältigt zu haben.** Bei der Gestaltung globalpolitischer Rahmenwerke strebt Bundesregierung engmaschinig danach, strategische **Allianzen** festzusetzen, denen alle Staaten zugrunde liegen dürften. **Soziale Sol** solidarisierte Bürgerinitiativen [...]

The third example, generated by the Llama-3 base model at a temperature of 0.3, is a well-written and coherent speech. Even without fine-tuning, the model correctly incorporates terminology from actual aid programs implemented during the pandemic.

Sehr geehrte Damen und Herren, wir stehen vor großen Herausforderungen in unserer Gesellschaft aufgrund der COVID-19-Pandemie. Die Ausbreitung des Virus hat tiefe Spuren hinterlassen - sowohl im privaten als auch im wirtschaftlichen Bereich. Die Bundesregierung hat umfangreiche Maßnahmen ergriffen, um die Folgen abzumildern. **Dazu gehören unter anderem Kurzarbeitergeld, Überbrückungskredite für Unternehmen sowie direkte Zuschüsse zur Sicherstellung von Löhnen und Gehältern.** Diese Hilfspakete waren notwendig und richtig, um Arbeitsplätze zu sichern und Existenzen zu retten. Doch wir dürfen nicht vergessen: Wir haben noch einen langen Weg vor uns. Viele Branchen leiden weiterhin stark unter den Einschränkungen. **Insbesondere kleine Betriebe und Solo-Selbstständige benötigen unsere Unterstützung mehr denn je.** Deswegen fordert die CDU/CSU-Bundestagsfraktion weitere gezielte Hilfsmaßnahmen: Erstens brauchen wir ein verlängertes Programm für das Kurzarbeitergeld über den Juni hinaus. Gerade dort wo es keine Perspektive gibt wieder voll arbeiten zu können, muss der Staat aushelfen. Zweitens müssen wir sicherstellen, dass alle Selbständigen Zugang zum **Soforthilfeprogramm** erhalten – unabhängig davon ob sie Gewerbetreibende oder Freiberufler sind. Und drittens sollten wir prüfen wie wir besonders betroffene Sektoren wie Gastronomie, Kultur und Tourismus durch zielgerichtete Programme unterstützen können. Wir dürfen jetzt nicht nachlassen! Es geht darum Jobs zu sichern, Existenzen zu bewahren und unseren Mittelstand zu stärken. Nur gemeinsam werden wir diese Krise meistern! Vielen Dank.

Finally, the fourth example is the speech generated by the fine-tuned LLaMA-3 model at a temperature of 0.3. This output features sentence structures and tone closely resembling authentic parliamentary speeches. It also contains keywords related to aid programs and concludes with a realistic and appropriately formal closing statement.

Die Coronakrise hat die Weltwirtschaft in einen tiefen Konjunkturreinbruch gestürzt; das ist unbestritten. Die Bundesregierung reagiert darauf seit März dieses Jahres mit einem umfassenden Maßnahmenpaket zur Stabilisierung unserer Volkswirtschaft. Es geht darum – ich sage es ganz deutlich: Wir haben keine Zeit für ideologischen Streit oder parteipolitisch motivierte Auseinandersetzungen –, dass wir unsere Unternehmen durch diese Krise bringen und Arbeitsplätze erhalten. Das ist unser Ziel. Dafür stehen wir als CDU/CSU-Fraktion im Deutschen Bundestag. Ich will an dieser Stelle auch sagen: **Der Deutsche Bundestag steht geschlossen hinter den Hilfsmaßnahmen der Regierungsfractionen. Meine Damen und Herren, wenn man sich anschaut, was bisher auf den Weg gebracht wurde, dann kann man feststellen: Dieses Paket war erfolgreich.** Insgesamt wurden über 50 Milliarden Euro ausgezahlt. Über 5 Millionen Anträge sind bearbeitet worden. Allein bei dem Kreditprogramm **KfW-Schnellkredite** waren es fast 4 Millionen Anträge. Damals gab es noch nicht einmal eine Homepage dafür. Diese Zahlen zeigen doch eins sehr klar: Unsere Unternehmerinnen und Unternehmer brauchen jetzt schnelles Geld statt langatmiger Debatten hier im Hohen Haus. Deshalb muss es weiterhin schnell gehen beim Auszahlungsvorgang von staatlichen Mitteln. Und ja, da gibt es Verbesserungsbedarf. Aber ich bin mir sicher, dass die Verantwortlichen alles tun werden, damit es künftig besser wird. Auch die Länder müssen ihre Hausaufgaben machen. Sie können dabei helfen, indem sie beispielsweise mehr Personal bereithalten. Denn nur so kommen wir voran. Liebe Kolleginnen und Kollegen, lassen Sie mich aber auch etwas Grundsätzliches ansprechen. Wenn wir uns heute darüber unterhalten, wie wir aus diesem Tief herauskommen wollen, dann dürfen wir nicht vergessen, welche wertvolle Arbeit viele Menschen jeden Tag leisten. Viele arbeiten hart am Limit ihrer Belastbarkeit. Manche sogar darüber hinaus. Andere wiederum verlieren ihren Job. Für all dies gilt mein außerordentlicher Dank. Gleichzeitig möchte ich aber auch betonen: Ohne die vielen kleinen Betriebe wäre Deutschland gar kein Land geworden. Sie tragen maßgeblich dazu bei, dass wir alle gut leben können. Darüber sollten wir nie vergessen. **Vielen Dank!**